

MEMORANDUM

DATE	September 23, 2026
RE	Do the structural merger screens predict realised price effects? Evidence from five US airline mergers
DATA	DOT Origin and Destination Survey (DB1B), 2005Q1-2019Q4; 344 million ticket records

SUMMARY

Across the five large US airline mergers that closed between 2008 and 2016, fares on routes where the merging carriers overlapped rose 1.3% relative to matched control routes, and 2.3% once the carriers were operating under a single certificate. The effect is absent for the first four quarters after closing and emerges only afterwards, which is when fare systems actually combine.

The structural screen in Section 2.1 of the 2023 Merger Guidelines does not identify which of those routes the increases landed on. Sorting overlap routes into deciles by the increase in HHI produces a flat profile of realised fare effects: +2.3% in the bottom decile against +1.8% in the top. Routes the presumption flags saw fare changes 0.71 points lower than routes it does not flag, not higher.

Diversion-based measures do identify them. The same routes sorted by GUPPI climb monotonically from -1.9% in the bottom decile to +7.9% in the top. A one-standard-deviation increase in GUPPI is associated with a 2.77 point larger realised fare increase (standard error 0.22); the same figure for the HHI increase is 0.23 (0.20), which cannot be told from zero.

WHAT WAS DONE

A route is an overlap route for a merger if both carriers each held at least 1% of its passengers in the four quarters before the deal was announced. Controls are routes where exactly one of the two was selling, matched to overlap routes on distance, traffic, concentration, fare, nonstop share and pre-merger fare trend. Both groups absorb whatever the merger did to the combined carrier's costs and network; only overlap routes lose a competitor.

Effects are estimated by stacked difference-in-differences, each merger in its own window against its own controls. The pooled two-way fixed effects estimator is reported for contrast and not relied on: a Goodman-Bacon decomposition shows 14% of it rests on comparisons that use already-merged routes as the control group for later deals, and it returns -2.4%, of the wrong sign.

Predictions were computed from pre-announcement information only: HHI and its increase from baseline shares, and GUPPI, diversion and simulated fare effects from a Bertrand model with logit demand, solved route by route.

HOW MUCH TO TRUST IT

The design passes its placebo: routes where only one merging carrier flew, compared against routes where neither did, show -0.04% [-0.98, +0.90], so the estimator is not picking up system-wide effects and calling them competitive ones. Pre-merger coefficients are individually indistinguishable from zero. A wild cluster bootstrap gives $p = 0.003$.

Two limits matter. First, the demand system could not be estimated credibly: four of the five candidate instruments return a price coefficient of the wrong sign, because entry responds to demand, and the one valid cost instrument implies inelastic firm-level demand that no profit-maximising carrier could be pricing against. The price coefficient used in the simulation is therefore calibrated to a target elasticity of -2.5, not estimated. Second, that calibration sets the level of every GUPPI and every simulated price change, but it barely moves their ordering across routes - the rank correlation across alternative calibrations does not fall below 0.92. The finding above is about ordering, which is the part that survives.

WHAT FOLLOWS

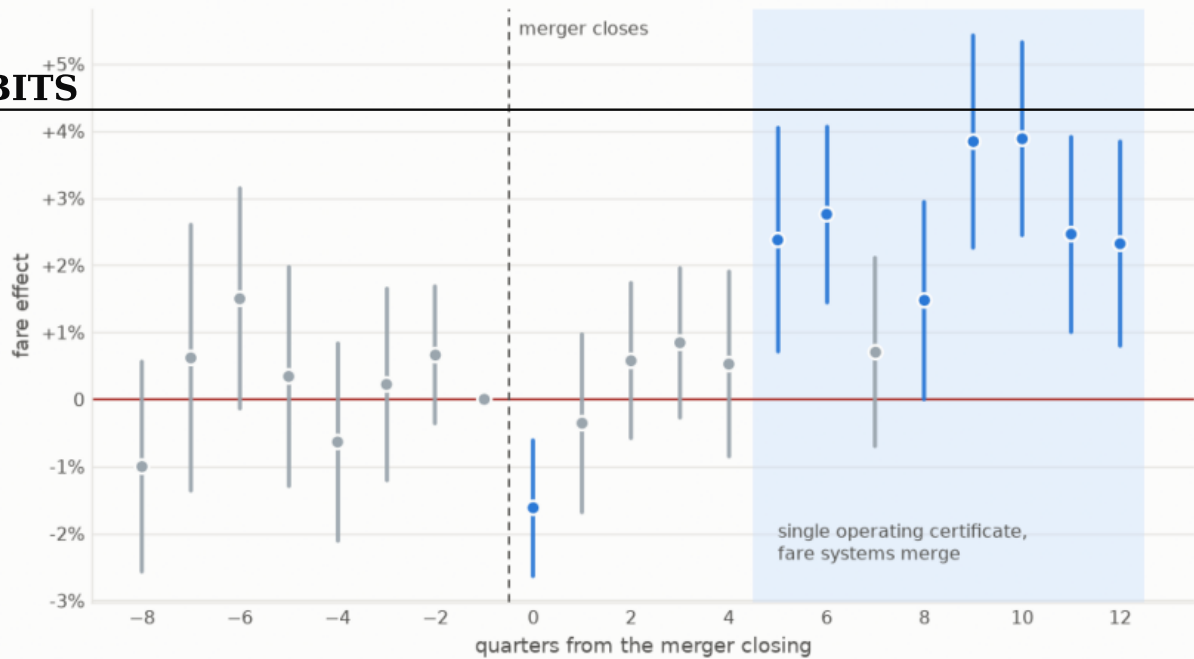
On this evidence, moving the HHI threshold would not help: at every candidate cut-off from 50 to 2,000 points the flagged group has a smaller average realised fare increase than the unflagged group. The problem is not where the line sits but that concentration and realised harm are close to uncorrelated here, once overlap is already established. A diversion-based screen ran on the same pre-merger data and sorted the same routes correctly.

That comparison is available before a decision has to be made, and this retrospective is the kind of exercise that establishes whether it is worth the extra work. It is one industry and five deals, and airline networks have features - hub economics, connecting itineraries, frequent-flyer lock-in - that may not generalise. The method transfers even where the magnitudes do not.

EXHIBITS

Fares on overlap routes rise, but only after the airlines actually merge

Effect on the average fare of an overlap route relative to matched control routes, by quarter from closing. Grey where the 95% interval covers zero. The quarter before closing is the omitted baseline.



DOT DB1B, 2005-2019. Stacked difference-in-differences, 7,556 routes, standard errors clustered on the route. Average over quarters +5 to +12: +2.28%.

Exhibit 1. The realised effect, quarter by quarter.

Do routes the tools rank as worse actually see bigger fare increases?

Overlap routes sorted into deciles by each prediction, with the average realised fare effect in each. Grey where the interval covers zero.

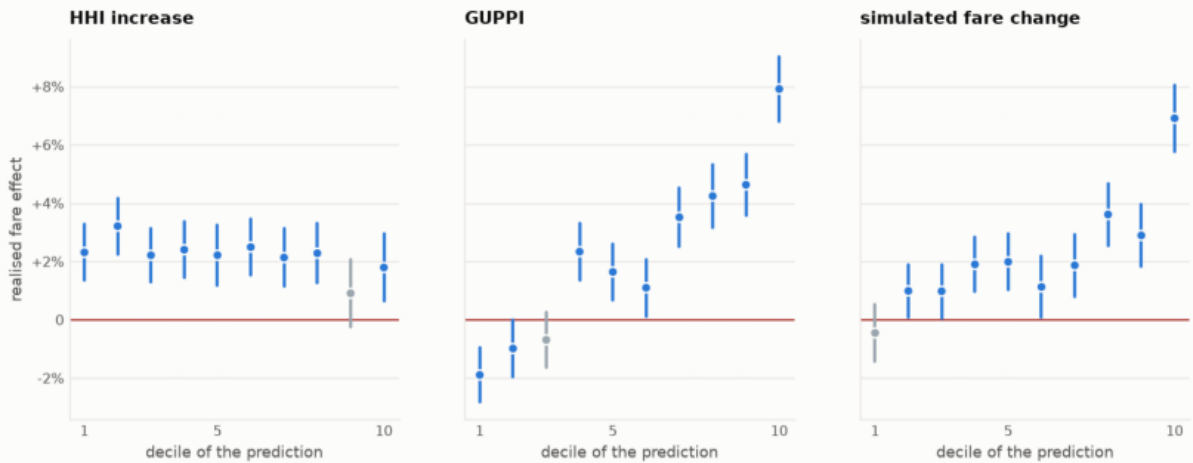


Exhibit 2. Realised effect by decile of each prediction.